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While we wait for the workshop to begin, please download all these !

plant health 2026 August 1-4 • Providence, Rhode Island

Please go this GitHub repository and download all the available files:
https://github.com/Potnislab/APS-Plant-Health-Workshop-2026

On the main page, all the links are present to install the required applications mentioned below:

- ✓ **Laptop with terminal access (mac) or putty (windows)**

  Windows > PuTTY

  > terminal

- ✓ **DUO mobile app installed in your cellphones**

- ✓ **igv app installed in laptop**

- ✓ **Filezilla app installed in laptop**

- ✓ **R / RStudio**

Genomics and Metagenomics Tools & Approaches

Revealing the Hidden Diversity in Agricultural Fields

Pre-Meeting Workshop
August 1st, 2026 (8 AM to 12 PM)

Presented By:

Palash Ghosh (Part 1: Variant Calling)
Amanpreet Kaur (Part 2: Metagenomics)
Auburn University

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We Are

Welcome Everyone!

Palash Ghosh
(pzg0041@auburn.edu)

Amanpreet Kaur
(azk0151@auburn.edu)

PhD Candidates in POTNIS Lab
Auburn University, AL.

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Introductions

Welcome Everyone!
Please take a moment to introduce yourself to the group:

1. Name & Affiliation

- Tell us your name, your institution/organization, and your current lab or research group.

2. Research Focus

- What crop systems, plant pathogens, or biological organisms are you currently studying?

3. Reason for Interest

- What motivated you to join this workshop?

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Workshop Objectives

Genomics and Metagenomics Tools & Approaches: Revealing Hidden Diversity in Agricultural Fields

SESSION OBJECTIVES

- **Selection of Analytical Tools:** Identify and match specific phytopathology research questions to appropriate bioinformatic packages, from genomics to metagenomics pipelines.
- **Assess Pathogen Diversity:** Execute robust methods for variant calling, strain profiling, and population structure analyses to evaluate pathogen diversity in field samples.
- **Build Reproducible Workflows:** Confidently adapt and implement the provided code templates and pipelines to analyze their own genomic and metagenomic datasets.

MATERIALS & RESOURCES PROVIDED:

✓ Bash scripts (all pipeline steps) with example datasets (*Xanthomonas*)
✓ Complete workflows with Step-by-Step PDFs (Handouts)
✓ GitHub repositories

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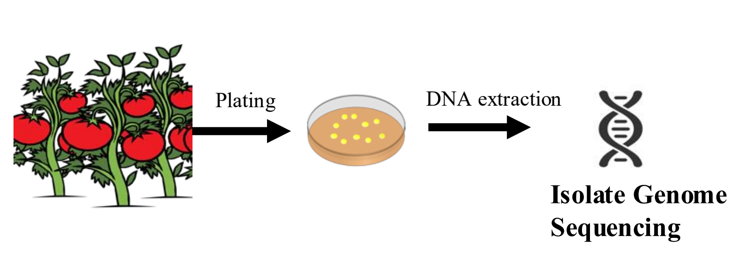
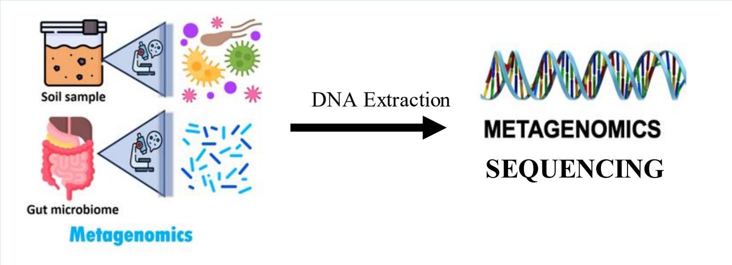
Genomics & Metagenomics

GENOMICS



METAGENOMICS



Workshop Roadmap:

What we will do

GENOMICS

Using single genome isolates (Part 1)

✓ **Read Mapping and Quality Control:** Aligning paired-end reads to reference & removing PCR duplicates *(BWA-MEM, Picard)*.

✓ **Variant Calling and Filtering:** Identify high-confidence SNPs/INDELs across isolates *(SAMtools, GATK)*.

✓ **Applications: Phylogenomics:** Resolve fine-scale evolutionary relationships & population structures *(Parsnp, RheirBaps)*.

METAGENOMICS

Using the mixed genome/field samples (Part 2)

✓ **Reference Database Construction:** Curate lineage-specific SNP signature from known species *(StrainEst)*.

✓ **Metagenomic Read Mapping:** Align mixed field reads directly to multi-strain database *(Bowtie2, SAMtools)*.

✓ **Strain Profiling & Abundance:** Use regression-based tool to quantify inter/intra-species diversity *(StrainEst)*.

While we wait for the workshop to begin, please download all these !

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Application_installation_instructions.docx should have all the links to the required applications mentioned below:

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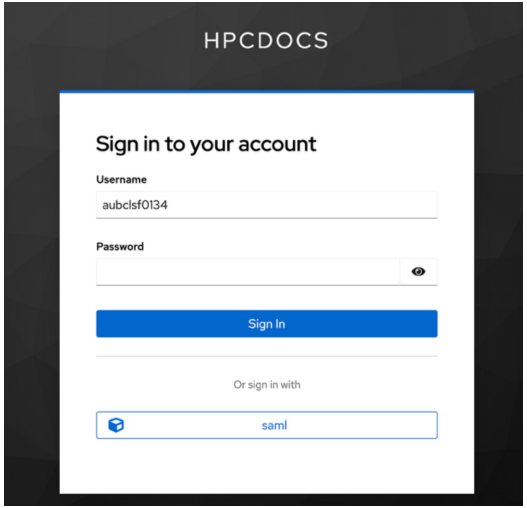
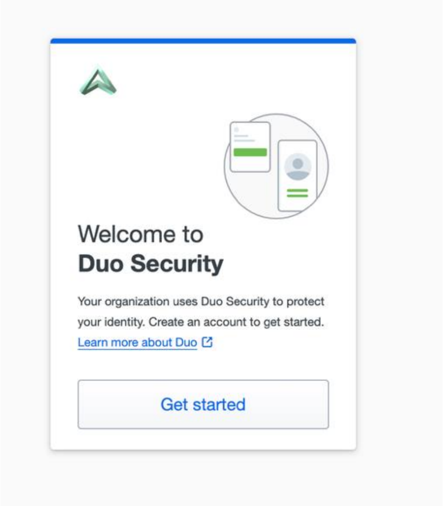
- ✓ **Filezilla app installed in laptop**

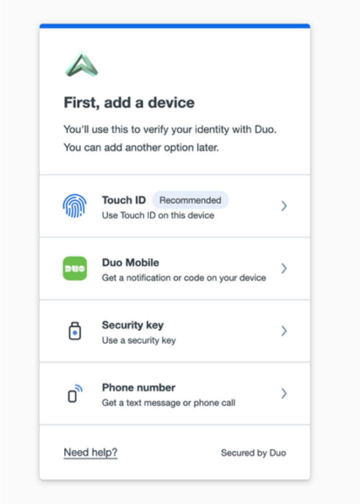
- ✓ **R / RStudio**

Enrolling in Duo-push

Step 1) By connecting to https://hpcdocs.asc.edu/ then following the prompts after you put in your username and temporary password.



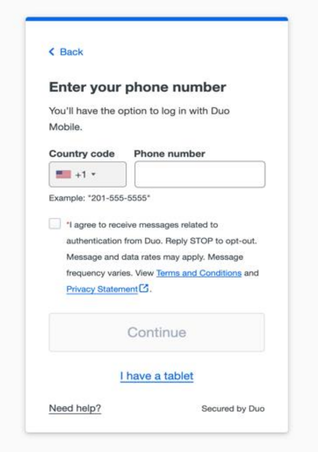


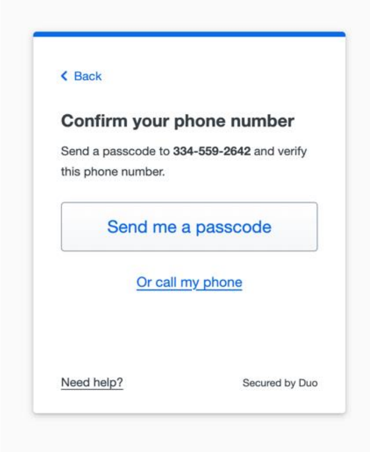


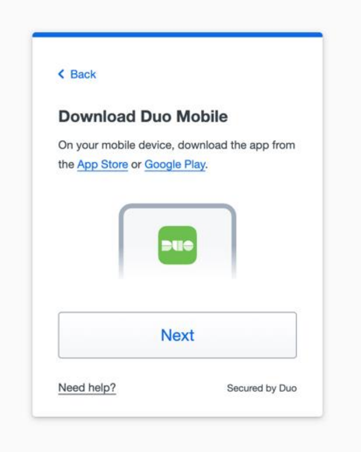
https://hpcdocs.asc.edu/

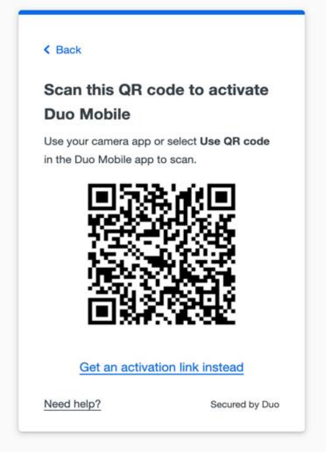
Enrolling in Duo-push

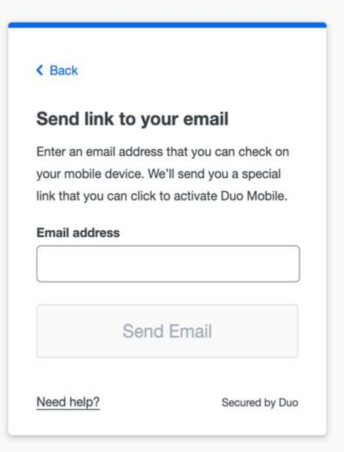
Step 2) Next you will get a dialog that lets you select the type of device (mobile phone, tablet, or landline). Select the 'Mobile phone' option and click 'Continue'.
Step 3) In the next dialog, enter your 10-digit phone number, check the confirmation box that appears, and click 'Continue'.
Step 4) DUO will attempt to auto sense what type of phone you have. If it can not automatically determine that, you will get a dialog like this one. Select the appropriate phone type and click 'Continue'. (You may NOT get this dialog. If not, go on to the next step.)





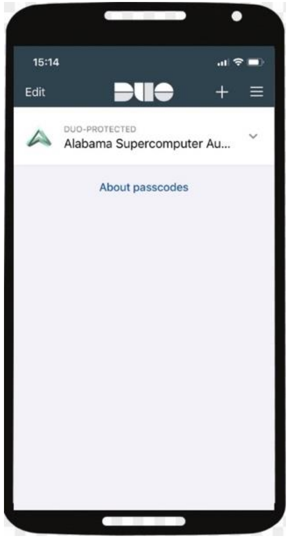


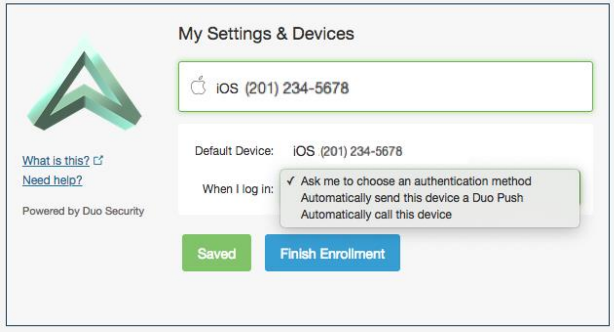




Enrolling in Duo-push

5) Install the Duo Mobile app via the app store on your smartphone, then click 'I have Duo Mobile installed'.
6) Open the Duo Mobile app on your phone, click '+' to add a new account, and scan the provided QR code. Click the 'Continue' button in your browser once it becomes available. The DUO app on your phone should now look something like this.
7) Select a default authentication method if desired and click 'Finish Enrollment'. The dialog will look like this.





Accessing the supercomputers

Temporary Class Account

WARNING: This is a temporary account. The account and all the files in it will be deleted at the end of the semester.

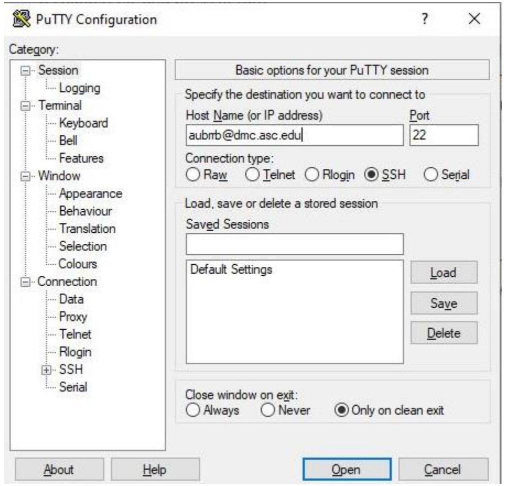
Pre-requisite: Putty Installed in Windows users and terminal is already present ready to use for macbook users.

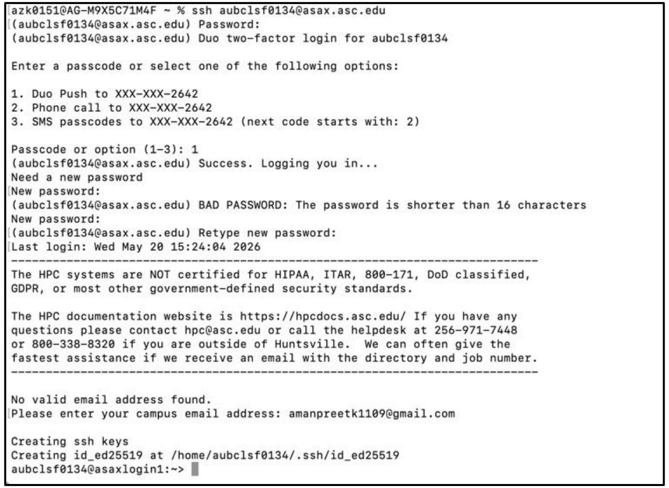
Step 1. Open terminal/Putty

Windows > PuTTY

> terminal

Step 2. Enter temporary username and password. It will give duo push and ask for new password





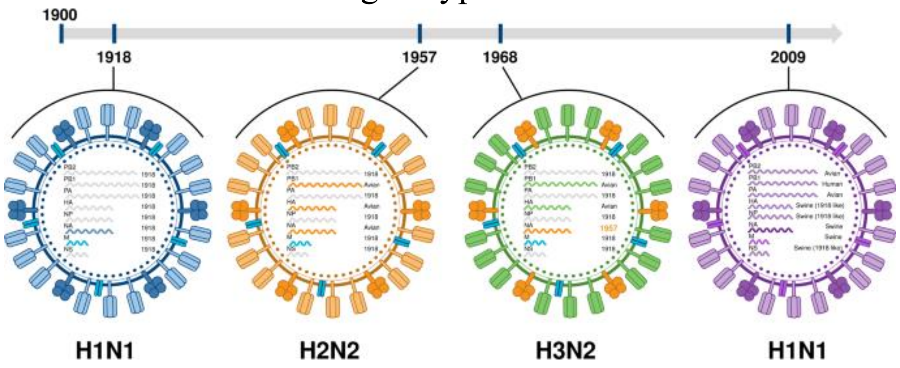
```
azk0151@AG-M9X5C71M4F ~ % ssh aubclsf0134@asax.asc.edu
(aubclsf0134@asax.asc.edu) Password:
(aubclsf0134@asax.asc.edu) Duo two-factor login for aubclsf0134

Enter a passcode or select one of the following options:

1. Duo Push to XXX-XXX-2642
2. Phone call to XXX-XXX-2642
3. SMS passcodes to XXX-XXX-2642 (next code starts with: 2)

Passcode or option (1-3): 1
(aubclsf0134@asax.asc.edu) Success. Logging you in...
Need a new password
New password:
(aubclsf0134@asax.asc.edu) BAD PASSWORD: The password is shorter than 16 characters
New password:
(aubclsf0134@asax.asc.edu) Retype new password:
Last login: Wed May 20 15:24:04 2026
--------------------------------------------------------------------------------
The HPC systems are NOT certified for HIPAA, ITAR, 800-171, DoD classified,
GDPR, or most other government-defined security standards.

The HPC documentation website is https://hpcdocs.asc.edu/ If you have any
questions please contact hpc@asc.edu or call the helpdesk at 256-971-7448
or 800-338-8320 if you are outside of Huntsville.  We can often give the
fastest assistance if we receive an email with the directory and job number.
--------------------------------------------------------------------------------

No valid email address found.
Please enter your campus email address: amanpreetk1109@gmail.com

Creating ssh keys
Creating id_ed25519 at /home/aubclsf0134/.ssh/id_ed25519
aubclsf0134@asaxlogin1:~>
```

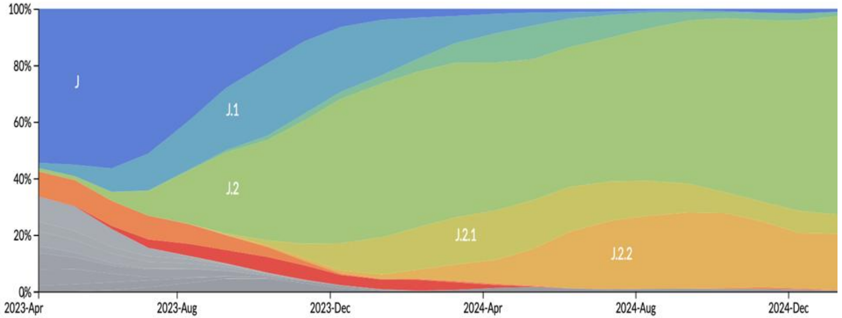
Part 1a.
Variant Calling

Xanthomonas perforans: Genomic Evolution Across Decades

WHAT IS VARIANT?

Variants/genotypes of Influenza Virus





Importance to Study Pathogen Diversity?

Pathogen diversity refers to the genetic variation and differences that exist within and between pathogenic populations

Why to Study It?

Studying pathogen diversity allows us to uncover the fundamental biological mechanisms driving evolution in the field.

- **Evolutionary Tracking:** Measure mutation rates and identify genomic recombination events.
- **Molecular Mechanisms:** Understand how surface proteins mutate to bypass host immunity.
- **Gene Flow:** Map horizontal gene transfer and mobile genetic elements (plasmids).

Why is it Important?

Understanding these genetic variations is critical for protecting global food security and developing sustainable agriculture.

- **Disease Management:** Anticipate outbreaks and prevent catastrophic agricultural yield losses.
- **Resistance Breeding:** Guide the development of durable, disease-resistant crop varieties.
- Track and mitigate the rise of antimicrobial/pesticide resistance.

WHAT IS VARIANT CALLING?

The process of identifying positions in a genome where a DNA sample differs from a reference sequence

How it works:

Reference Genome (your database):

A	T	G	C	A	T	G

Your Sample (what you sequenced):

A	T	G	G	A	T	G

DIFFERENCE FOUND!

Types of Variants We Detect:

SNP (Single Nucleotide Polymorphism)

One base changed

Ref: ...ATGC...

Alt: ...ATAC...

(G→A at position 3)

INDEL (Insertion/Deletion)

Bases added or removed

Ref: ...ATGCTA...

Alt: ...ATG-TA......

(C deleted)

Dataset to study:

Variants in *Xanthomonas perforans*

Genomic Evolution Across Decades

Our study focuses on identifying and characterizing mutations in *Xanthomonas perforans* across field-collected isolates spanning from 1991 to 2026.

- **Baseline Comparison:** Aligning modern isolates against the 1991 reference genome.
- **Temporal Tracking**
- Understanding field adaptation and identifying fitness-linked mutations.

Reference Genome 1991 → **1995** | **2000** | **2005** | **2010** | **2017**

To make the exercise simpler we have selected a smaller dataset of 5 representative isolates.

Assembly Accession	SCs_NEW	BioSample ID	Assembly Name	Isolation Date
GCA_041643375.1	SC1	SAMN33607657	ASM4164337v1	1991
GCA_041643575.1	SC1	SAMN33607647	ASM4164357v1	1995
GCA_041643495.1	SC7	SAMN33607652	ASM4164349v1	1998
GCA_041643415.1	SC1	SAMN33607656	ASM4164341v1	2006
GCA_041646435.1	SC7	SAMN33607572	ASM4164643v1	2011
GCA_019987715.1	SC3	SAMN16476190	ASM1998771v1	2017

And we will use illumina short reads for the workshop

Learning Objectives

- Master read mapping algorithms and BWA-MEM
- Execute complete variant calling pipelines
- Implement quality control & duplicate removal
- Interpret VCF files and annotations
- Apply confidence-based variant filtering
- Analyze temporal genomic evolution

STEP 1: Reference Genome Indexing

Create searchable genome index to reduce mapping time from hours to minutes

BWA Index

Burrows-Wheeler Transform

FASTA Index

Coordinates for genome access

Dictionary

GATK requirement

STEP 2: Read Mapping with BWA-MEM

Why BWA-MEM?

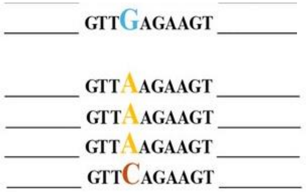
- 70–1 Mb reads
- High accuracy
- Split alignments
- Industry standard

Output: SAM File

CHROM	POS	REF	ALT	QUAL

→ *Converted to BAM (binary format)*

Paired-end reads aligned to 1991 reference genome



Reference (*Xp* genome of 1991)

Sample Reads

Read Mapping

STEP 3: SAM → BAM Conversion & Sorting

SAM

✗ Human-readable

✗ Large files

✗ Slow parsing

→

BAM

✓ Binary format

✓ 5-10× smaller

✓ Fast indexed access

STEP 4: PCR Duplicate Removal

PCR amplification creates duplicate reads → False positive variants → Must remove

Identify

Same start coordinates

Evaluate

Check quality scores

Keep

Highest quality only

STEP 5: Variant Calling with bcftools

Bayesian statistics + pileup data → Detect SNPs & INDELs

CHROM Chromosome	**POS** Position	**REF/ALT** Alleles
QUAL Quality	**DP** Depth	**MQ** Mapping Quality

STEP 6: Variant Filtering

Using **bcftools** (Bayesian statistics + pileup data) to detect SNPs, followed by strict confidence-based filtering protocols to ensure data integrity.

Filtering Parameter	Threshold	Scientific Rationale
Read Depth (DP)	**> 15**	Ensures sufficient coverage across the genome to support the variant call over background sequencing noise.
Mapping Quality (MQ)	**> 50**	Guarantees the read is aligned to the correct region, eliminating ambiguous multi-mapped reads.
Confidence Score (QUAL)	**> 100**	High PHRED-scaled probability that the detected polymorphism is a genuine biological mutation.
Zygosity (FQ)	**< 0.025**	Targets homozygous (fixed) mutations within the haploid bacterial population, filtering out transient errors.

Expected Outputs

1

Indexed Genome

2

Aligned BAMs

3

Raw VCF

4

Filtered SNPs

5

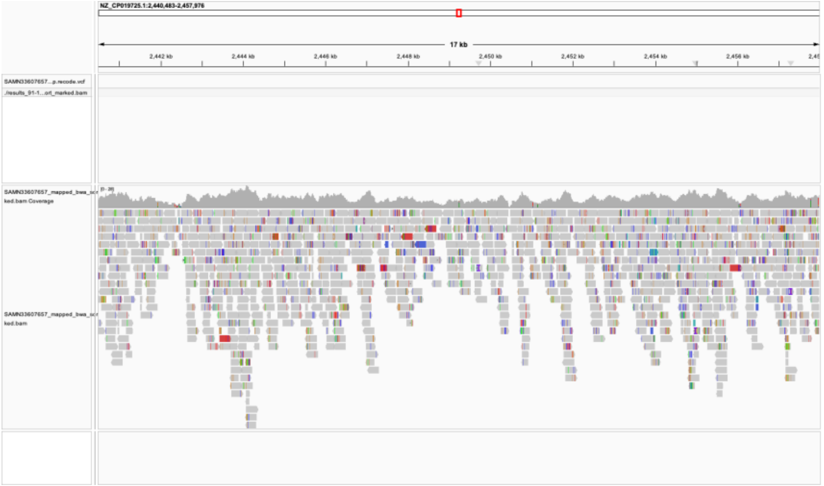
Mutation Tally

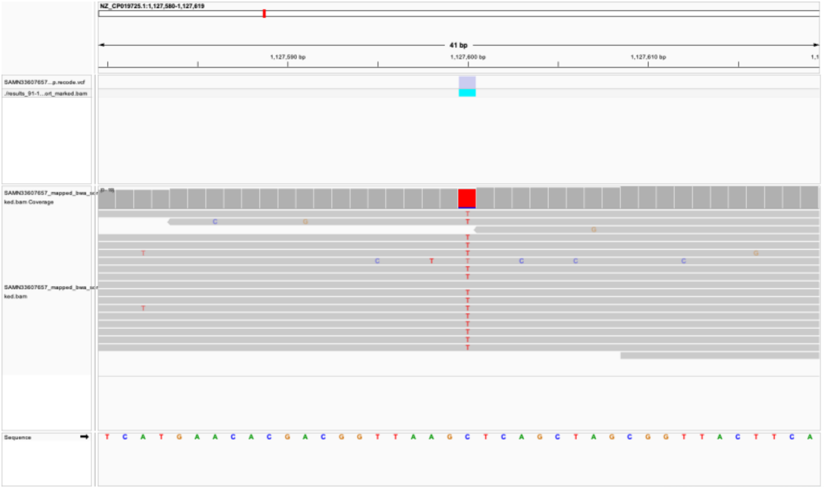
OUTPUTS

```
After filtering, kept 1 out of 1 Individuals
Outputting VCF file...
After filtering, kept 0 out of a possible 703 Sites
No data left for analysis!
Run Time = 0.00 seconds
Finished processing SAMN33607657
==================================================
FINAL MUTATION COUNTS (Compared to Reference)
==================================================
SAMN16476190 : 6872 total SNPs
SAMN33607572 : 3665 total SNPs
SAMN33607647 : 70 total SNPs
SAMN33607652 : 147 total SNPs
SAMN33607656 : 2029 total SNPs
SAMN33607657 : 0 total SNPs
==================================================
PIPELINE COMPLETE
==================================================
```

OUTPUTS

The final output generates high-quality VCF files and a precise mutation tally. These filtered SNPs are visualized (e.g., via IGV) to reveal genomic signatures of field adaptation.





Let’s discuss potential applications!

- What are the potential applications of variant calling?

- How would you apply these tools to your own research?

Discuss in your groups for 5 mins, then nominate one person to share your ideas with the room

COFFEE BREAK (~20 min)

Part 1b.
Applications of Variant Calling

Population Structure: Phylogenomics

Using Core-Genome SNP Analysis for Bacterial Population Evolution

1. **Data Acquisition**
   Download complete *Xanthomonas perforans* genomes from NCBI GenBank

2. **Genome Annotation**
   Use Prokka to annotate genomes in GBK format (Reference requirement for Parsnp)

3. **Core-Genome Alignment**
   Use Parsnp for alignment against reference genome

4. **SNP Extraction**
   Extract SNPs with HarvestTools into FASTA format

5. **Population Structure**
   Use RhierBaps to define lineages and evolutionary relationships

Phylogenomic tools & their roles

Parsnp

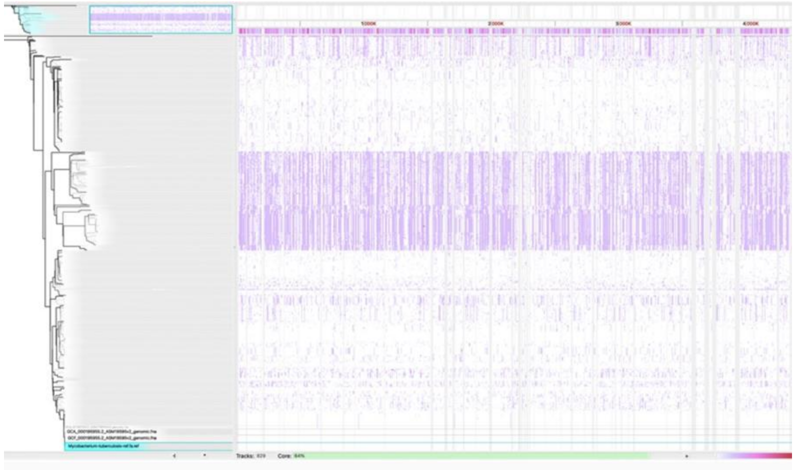
Core-Genome Alignment

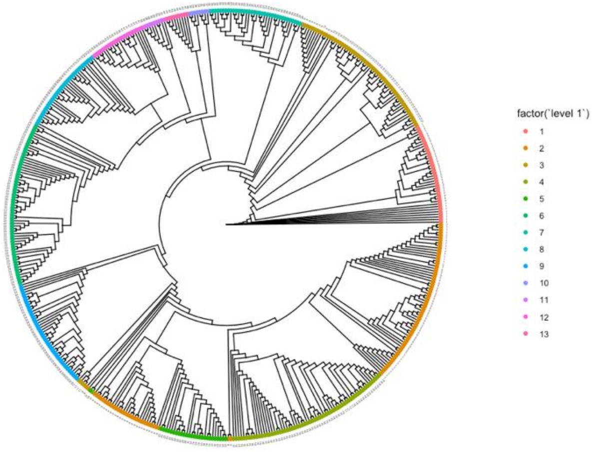
Aligns query genomes to reference, identifies core regions and SNPs

RhierBaps

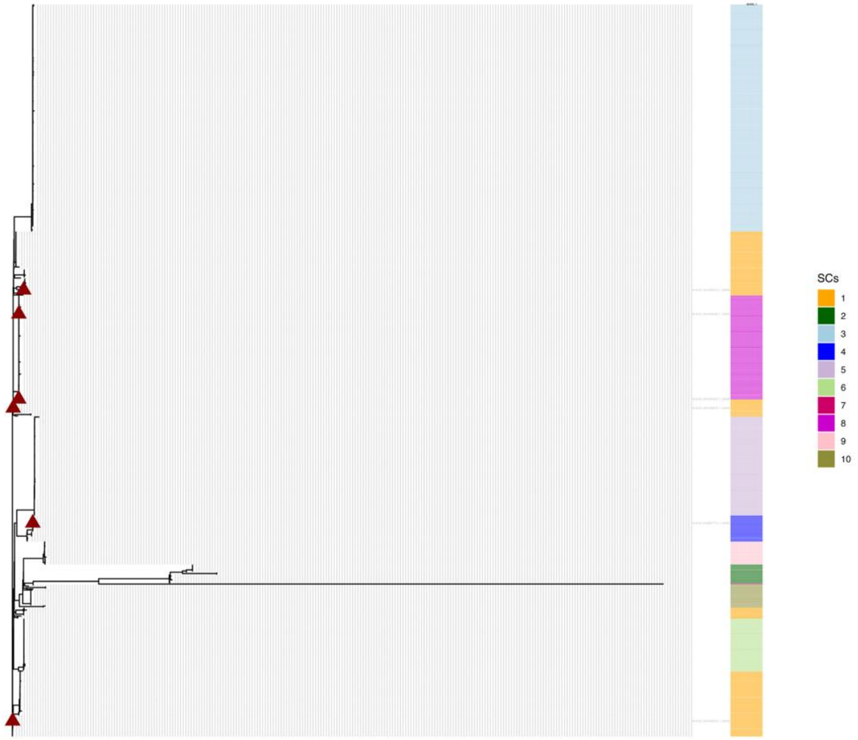
Population Structure

Hierarchical Bayesian clustering for lineage definition and visualization





Results



10 lineages of *Xp* populations has been emerged all over the world

Limitations

Reference Genome Bias

If reference is basal/distant/poor quality, core-regions underestimated; SNPs may be mis-assigned

✓ *Select representative reference; perform multiple alignments; use reference-free methods*

Core-genome Bias

Core-genome focuses on variable sites; rare/novel variants typically excluded

✓ *Use Pangenome analysis*

Assumes Vertical Transfer

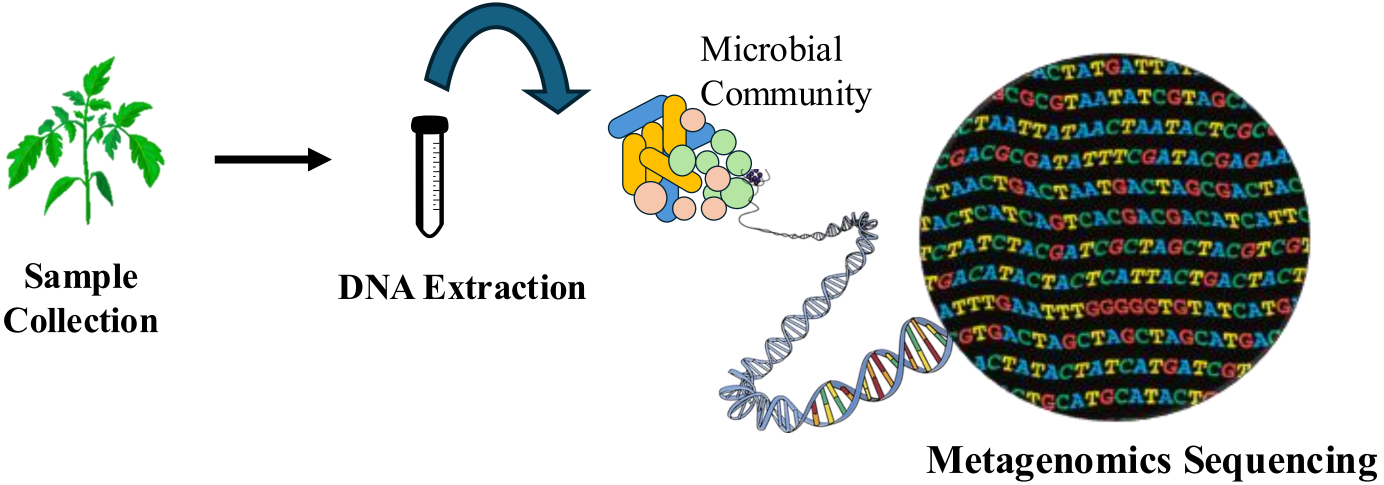
It could over-cluster the data

✓ *Run recombination test on your datasets before running this.*

BREAK (~10 min)

Part 2
Metagenomics & field diversity

Metagenomics



Using Strain-resolved Metagenomics

Field Study

Interaction between climatic variation and pathogen diversity shape endemic disease dynamics in the agricultural settings

Rishi Bhandari, Amanpreet Kaur, Ivory Russell, Ogonnaya Michael Romanus, Destiny Brokaw, Anthony P Keinath, Zachary Snipes, Philip Rollins, Nettie Baugher, Inga Meadows, Ty Nicholas Torrance, Bhabesh Dutta, Edward Sikora, Roberto Molinari, Samuel Soubeyrand, Neha Potnis

doi: https://doi.org/10.1101/2024.09.27.615185

Controlled Conditions

Pathogen

ORIGINAL ARTICLE | Open Access

Navigating Host Immunity and Concurrent Ozone Stress: Strain-Resolved Metagenomics Reveals Maintenance of Intraspecific Diversity and Genetic Variation in *Xanthomonas* on Pepper

Amanpreet Kaur, Ivory Russell, Ranlin Liu, Auston Holland, Rishi Bhandari, Neha Potnis

Microbes

Article | Open access | Published: 27 March 2023

***Xanthomonas* infection and ozone stress distinctly influence the microbial community structure and interactions in the pepper phyllosphere**

Rishi Bhandari, Alvaro Sanz-Saez, Courtney P. Leisner & Neha Potnis

Host

Physiological responses of pepper (Capsicum annum) to combined ozone and pathogen stress

Collin Modelski [1], Neha Potnis [2], Alvaro Sanz-Saez [3], Courtney P Leisner [1 4]

Affiliations + expand

PMID: 38924220 DOI: 10.1111/tpj.16888

Inter-Species vs Intra-Species Diversity

Inter-Species Diversity

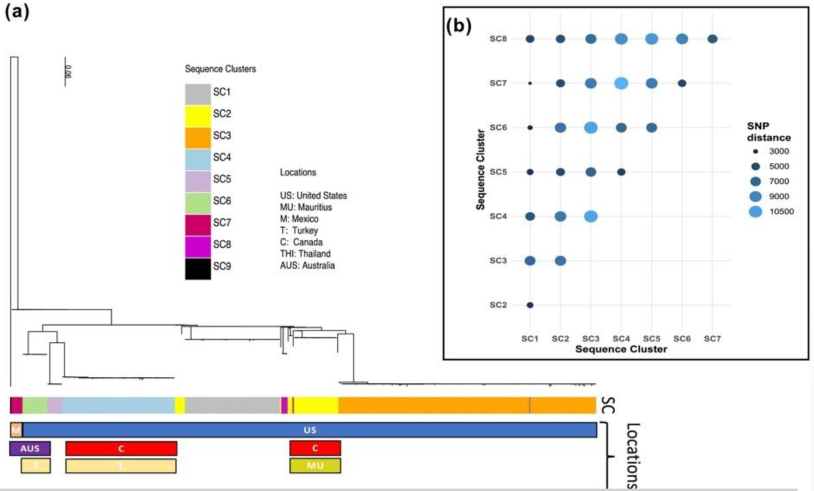
- Different bacterial species
- Major sequence divergence

- Example:
X. perforans,
X. euvesicatoria,
X. gardenari,
X. vesicatoria

Intra-Species Diversity

- Strains within same species
- ~99% sequence identity
- Distinct lineages/clones

- Example:
lineages of *X. perforans*



StrainEst Pipeline Overview

Phase 1: Reference Database Construction

- Curate database of reference genomes (known species/lineages)
- Extract SNPs that distinguish among references
- Group references into clusters based on SNP similarity
- Create lineage-specific SNP signature database

Phase 2: Metagenomic Sample Analysis

- Map metagenomic reads to reference database (Bowtie2)
- Quantify read coverage at SNP sites
- Use StrainEst (Lasso-based regression) to estimate strains
- Calculate relative abundances of each strain

StrainEst Tools & Approach

Bowtie2
(Read Mapping)

Fast, memory-efficient alignment of metagenomic reads to SNP reference database

Samtools
(Coverage Quantification)

Extract and quantify read coverage at SNP sites; filter by quality

StrainEst
(Strain Estimation)

Lasso-based regression to identify coexisting strains and estimate abundances

99% Similarity Threshold = 250-1,150 SNV Sites

Let's discuss potential applications!

- What are the potential applications and limitations of this pipeline?
- How would you apply these tools to your own research?

Discuss in your groups for 5 mins, then nominate one person to share your ideas with the room

StrainEst: Limitations & Considerations

Novel Strain Bias

Wild/novel strains missing from reference may cause overestimation of complexity or mapping failures

Sequencing Depth Bias

Low-frequency strains require high coverage; low-coverage positions must be filtered

Recombination Sensitivity

Relies on genetic linkage between marker SNPs; highly recombinogenic species need specialized methods

Strain Misidentification

Strains >99% identical with few SNP sites may fail to be distinguished reliably

Acknowledgments /Sponsors

plant health 2026

Bacteriology
Evolutionary and Genomics Communities

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College of Agriculture

POTNIS LAB

THANK YOU!